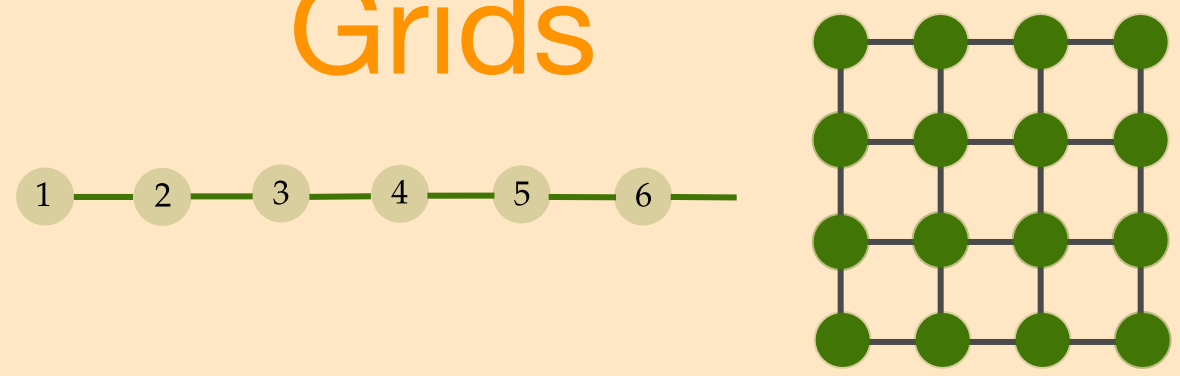


Data, domain, model

Scalar $\mathbb{R}$
Categorical $\mathbb{Z}$
Angle $\mathbb{S}$
high-dim data $\mathbb{R}^n$
3d (coordinates $\mathbb{R}^3$)
 m angles $\mathbb{T}^m$

Grids
1 2 3 4 5 6
Graphs
SCs
Ellipsoid
Sphere
1D time
2D/3D surface



Geometric models: EGNN, GAT, TFN ...

Topological models: GNN, SNN ...

Overview

Convolution

- Simplicial Fourier transform, Frequency, Fourier basis
- Convolutional filters
- Generalizes graph convolutions

Gaussian Process

- How to define GPs for different parts?
- Hodge-compositional idea
- Probabilistic methods

SC $\sim$ Hodge decomposition

Convolutional NNs

- Architecture (attention, message passing)
- Equivariance, Robustness
- Higher-order link predictions
- Generalizes GCN, Chebnet, etc.

Generative learning

- (Dynamic) optimal transport (Schrödinger bridge) on SCs
- Gaussian bridge
- Generative models: diffusion, flow models on SCs